

This Appendix gives a definitive summary of the complete *Mathematica* system. Most of what it contains you will never need to know for any particular application of *Mathematica*.

You should realize that this Appendix describes *all* the features of *Mathematica*, independent of their importance in typical usage.

Other parts of this book are organized along pedagogical lines, emphasizing important points, and giving details only when they are needed.

This Appendix gives all the details of every feature. As a result, you will often find obscure details discussed alongside very common and important functions. Just remember that this Appendix is intended for *reference purposes*, not for sequential reading. Do not be put off by the complexity of some of what you see; you will almost certainly never have to use it. But if you do end up having to use it, you will probably be happy that it is there.

By experimenting with *Mathematica*, you may find features that go beyond what is described in this Appendix. You really should not use any such features: there is no guarantee that features which are not documented will continue to be supported in future versions of *Mathematica*.

About the illustration overleaf:

This is a parametric plot of a sphere, made up from a sequence of points at equally-spaced parameter values. The command used was `PointParametricPlot3D[list, urange, vrange]`, where *list* is `{Sin[u] Cos[v], Sin[u] Sin[v], Cos[u]}`, *urange* is `{u, 0, Pi, Pi/15}`, and *vrange* is `{v, 0, 2Pi, Pi/30}`. The option setting `BoxRatios->{1,1,1}` was used.